Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let $f(x)=\begin{cases} 3x-4, & 0 \leq x \leq 2 \\ 2x+\ell, & 2<x \leq 9 \end{cases}$

If f is continuous at $\mathrm{x}=2$, then what is the value of ℓ?

(A) 0

(B) 2

(C) -2

(D) -1

Q.2

If $f(x)=(x+1)^{\cot x}$ is continuous at $x=0$, then what is $\mathrm{f}(0)$ equal to?

(A) 1

(B) e

(C) $\frac{1}{\mathrm{e}}$

(D) e^2

Q.3

The number of points at which the function $f(x)=\frac{1}{\log_e|x|}$ is discontinuous, is

(A) 1

(B) 2

(C) 3

(D) 4

Q.4

If the function $f(x)=\begin{cases} \frac{k \cos x}{\pi-2x}, & \text{when } x \neq \frac{\pi}{2} \\ 3, & \text{when } x=\frac{\pi}{2} \end{cases}$ be continuous at $\mathrm{x}=\frac{\pi}{2}$, then $\mathrm{k}=$

(A) 3

(B) 6

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(C) 12

(D) None of these

Q.5

Which of the following function(s) has/have removable discontinuity at $x = 1$?

(A) $f(x) = \frac{1}{\ln |x|}$

(B) $f(x) = \frac{1}{x^3 - 1}$

(C) $f(x) = 2^{2^{\frac{1}{1-x}}}$

(D) $f(x) = \frac{\sqrt{x+1} - \sqrt{2x}}{x^2 - x}$

Q.6

What is $\lim_{x \to 0} \frac{\sqrt{1+x} - 1}{x}$ equal to ?

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) $-\frac{1}{2}$

Q.7

If $y = \frac{1}{t^2 + t - 2}$ where $t = \frac{1}{x-1}$, then find the number of points of discontinuities of $y = f(x)$,

(A) 1

(B) 2

(C) 3

(D) 4

Q.8

A function $f : R \to R$ is defined as $f(x) = x^2$ for $x \geq 0$ and $f(x) = -x$ for $x < 0$. Consider the following statements in respect of the above function :
1 The function is continuous at $x = 0$.
2 The function is differentiable at $x = 0$. Which of the above statements is/are correct?

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2

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Q.9

A function f is defined as follows $f(x) = x^p \cos\left(\frac{1}{x}\right), x \neq 0 f(0) = 0$
What conditions should be imposed on p so that f may be continuous at $x = 0$?

(A) $p = 0$

(B) $p > 0$

(C) $p < 0$

(D) No value of p

Q.10

If $f(x) = x + \frac{x}{1+x} + \frac{x}{(1+x)^2} + \ldots$ to ∞, then at
$x = 0, f(x)$

(A) has no limit

(B) is discontinuous

(C) is continuous but not differentiable

(D) is differentiable

Q.11

Let f be a continuous function on R such that $f(1/4n) = (\sin e^n)\, e^{-n^2} + \frac{n^2}{n^2+1}$
Then the value of $f(0)$ is

(A) 1

(B) $1/2$

(C) 0

(D) None of these

Q.12

Let $f''(x)$ be continuous at $x = 0$ and $f''(0) = 4$. Then value of $\lim_{x\to 0} \frac{2f(x)-3f(2x)+f(4x)}{x^2}$ is

(A) 12

(B) 10

(C) 6

(D) 4

Q.13

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If function $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 1-x, & \text{if } x \text{ is irrational} \end{cases}$, then the number of points at which $\mathrm{f}(\mathrm{x})$ is continuous, is -

(A) ∞

(B) 1

(C) 0

(D) None of these

Q.14

Consider the following in respect of the function $f(x) = \begin{cases} 2+x, & x \geq 0 \\ 2-x, & x < 0 \end{cases}$

1. $\lim f_{x \to 1}(x)$ does not exist.
2. $\mathrm{f}(\mathrm{x})$ is differentiable at $\mathrm{x} = 0$
3. $\mathrm{f}(\mathrm{x})$ is continuous at $\mathrm{x} = 0$

Which of the above statements is/are correct?

(A) 1 only

(B) 3 only

(C) 2 and 3 only

(D) 1 and 3 only

Q.15

If $fx = \begin{matrix} 2x^2+3 & ; & x>3 \\ ax^2+bx+1 & ; & x \leq 3 \end{matrix}$ is differentiable everywhere, then $\frac{a}{b^2}$ is equal to

(A) 5

(B) $\frac{7}{3}$

(C) 1

(D) $\frac{16}{9}$

Q.16

If $fx = \begin{matrix} \frac{e^{1+\frac{1}{x}} - a}{e^{\frac{1}{x}}+1} & : & x \neq 0 \\ b & : & x = 0 \end{matrix}$ (where a and b are arbitrary constants) is continuous at $x = 0$, then the value of a^2 is equal to
(use $e = 2.7$)

(A) 7.29

(B) 6.29

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(C) 5.25

(D) 8.25

Q.17

Let $fx = \begin{matrix} \frac{1 - \cos x}{2\pi - x^2} \frac{\sin^2 x}{\log 1 + 4\pi^2 - 4\pi x + x^2} & : & x \neq 2\pi \\ \lambda & : & x = 2\pi \end{matrix}$ is continuous at $x = 2\pi,$ then the value of λ is equal to

(A) 0.5

(B) 0.25

(C) 0.85

(D) 2.5

Q.18

The function $fx = \max 1 - x, \quad 1 + x, \quad 2 \quad \forall x \in R$ is

(A) discontinuous at exactly two points

(B) differentiable $\forall x \in R$

(C) differentiable $\forall x \in R - \{-1, 1\}$

(D) continuous $\forall x \in R - \{0, 1, -1\}$

Q.19

Let $\mathrm{f(x)} = \left[\mathrm{x}^3 - 3\right]$, $[\mathrm{x}] =$ G.I.F. Then the no. of points in the interval $(1, 2)$ where function is discontinuous is

(A) 5

(B) 4

(C) 6

(D) 3

Q.20

The function $f : \mathrm{R}\{0\} \to \mathrm{R}$ given by $f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$ can be made continuous at $x = 0$ by defining $f(0)$ as

(A) 0

(B) 1

(C) 2

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(D) -1

Answers & Solutions

Q.1 Answer:

-2

Solution:

Given function is :

$$f(x)=\begin{cases} 3x-4, & 0 \leq x \leq 2 \\ 2x+\ell, & 2<x \leq 9 \end{cases}$$

and also given that $\mathrm{f}(\mathrm{x})$ **is continuous at** $\mathrm{x}=2$ **For a function to be continuous at a point LHL** $=$ $\mathrm{RHL}=$ **V.F. at that point.** $\mathrm{f}(2)=2=\mathrm{V.F.}$

$\Rightarrow \mathrm{RHL}: \lim_{\mathrm{x} \to 2}(2\mathrm{x}+\ell)=3(2)-4$

$\Rightarrow \lim_{\mathrm{h} \to 0}\{2(2+\mathrm{h})+\ell\}=6-4$

$\Rightarrow 4+\ell=2, \Rightarrow \ell=-2$

Q.2 Answer:

e

Solution:

For a function to be continuous at a point the limit should exist and should be equal to the value of the function at that point. Here point is $\mathrm{x}=0$

$$\lim_{x \to 0} f(x)=\lim_{x \to 0}(x+1)^{\cot x}$$

and

$$=\lim_{x \to 0}(1+x)^{\cot x}=\lim_{x \to 0}(1+x)^{\frac{1}{x} \cdot x \cot x}$$

$$=\lim_{x \to 0}(1+x)^{\frac{1}{x} \lim_{x \to 0} \frac{x}{\tan x}}=\mathrm{e}^{1}=\mathrm{e}$$

Since limiting value of $\mathrm{f}(\mathrm{x})=\mathrm{e}$**, when** $\mathrm{x} \to 0, \mathrm{f}(0)$ **should also be equal to e.**

Q.3 Answer:

3

Solution:

The function $\log|x|$ **is not defined at** $x=0$ **So,** $x=0$ **is a point of discontinuity Also, for** $f(x)$ **to be defined;** $\log|x| \neq 0 \Rightarrow x \neq \pm 1$

Hence, $0, 1, -1$ **are three points of discontinuity.**

Q.4 Answer:

6

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Solution:

$f(\pi/2) = 3$. **Since** $f(\mathrm{x})$ **is continuous at** $\mathrm{x} = \pi/2 \Rightarrow \lim_{x\to\pi/2}\left(\frac{k\cos x}{\pi-2x}\right) = f\left(\frac{\pi}{2}\right) \Rightarrow \frac{k}{2} = 3 \Rightarrow k = 6$

Q.5 Answer:

$f(x) = \frac{\sqrt{x+1}-\sqrt{2x}}{x^2-x}$

Solution:

$\lim_{x\to1} f(x)$ **does not exist.**
(b) $\lim_{x\to1} f(x) =$ **does not exist.**
(c) $\lim_{x\to1} f(x)$ **does not exist.**
(d) $\lim_{x\to1} f(x) = \frac{-1}{2\sqrt{2}}$**, therefore** $f(x)$ **has removable discontinuity at** $x = 1$.

Q.6 Answer:

$\frac{1}{2}$

Solution:

$\lim_{x\to0} \frac{\sqrt{1+x}-1}{x}$
$= \lim_{x\to0} \frac{\sqrt{1+x}-1}{x} \times \frac{\sqrt{1+x}+1}{\sqrt{1+x}+1}$
$\lim_{x\to0} \frac{1+x-1}{x[\sqrt{1+x}+1]}$
$= \lim_{x\to0} \frac{1}{\sqrt{1+x}+1} = \frac{1}{\sqrt{1+0}+1} = \frac{1}{2}$

Q.7 Answer:

3

Solution:

$t = \frac{1}{x-1}$ **is discontinuous at** $x = 1$**. Also** $y = \frac{1}{t^2+t-2}$ **is discontinuous at** $t = -2$ **and** $t = 1$
When $t = -2, \frac{1}{x-1} = -2 \Rightarrow x = \frac{1}{2}$
When $t = 1, \frac{1}{x-1} \Rightarrow x = 2$
So, $y = f(x)$ **is discontinuous at three points** $x = 1, \frac{1}{2}, 2$

Q.8 Answer:

1 only

Solution:

$f : R \to R, f(x) = \begin{cases} x^2 & , \quad x \geq 0 \\ -x, & x < 0 \end{cases}$
For continuity at $x = 0\ f(0-0) = \lim_{h\to0} f(0-h)$
$= \lim_{h\to0}[(0-h)] = \lim_{h\to0} h = 0$

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$f(0+0) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0}(0+h)^2 = 0$
and $f(0) = 0$ $\mathrm{f(x)}$ **is continuous at** $\mathrm{x} = 0$
For differentiability at $x = 0$ $\lim_{h\to 0} \frac{-(-h)-0}{-h} = \lim_{h\to 0} = \frac{h}{-h} = -1$
and $\lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} h = 0$
$\mathrm{f(x)}$ **is not differentiable at** $\mathrm{x} - 0$

Q.9 Answer:

$\mathrm{p} > 0$

Solution:

Given function is defined as :

$$f(x) = \begin{cases} x^p \cos\left(\frac{1}{x}\right) & x \neq 0 \\ 0, & x = 0 \end{cases}$$

For continuity:
LHS : $\lim_{x\to 0} f(x) =$ **RHS** $\lim_{x\to 0} f(x) = f(0)$
$\Rightarrow \lim_{\mathrm{x}\to 0} \mathrm{f(x)} = \lim_{\mathrm{x}\to 0} \mathrm{x^p} \cos\left(\frac{1}{\mathrm{x}}\right) = 0$
$\Rightarrow \lim_{x\to 0} x^p \cos\left(\frac{1}{x}\right) = 0$
$\cos\left(\frac{1}{x}\right)$ **is always a finite quantity if** $x \to 0 \Rightarrow \quad x^{\mathrm{p}} = 0$
which is possible only if $\mathrm{p} > 0$

Q.10 Answer:

is discontinuous

Solution:

For $x \neq 0$**, we have,** $f(x) = x + \frac{x/1+x}{1-\frac{1}{1+x}} = x + \frac{x/1+x}{x/1+x} = x + 1$
For $x = 0, f(x) = 0$**. Thus,** $f(x) = \begin{cases} x+1, & x \neq 0 \\ 0, & x = 0 \end{cases}$
Clearly, $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = 1 \neq f(0)$
So, $\mathrm{f(x)}$ **is discontinuous and hence not differentiable at** $\mathrm{x} = 0.$

Q.11 Answer:

1

Solution:

As f **is continuous,** $f(0) = \lim_{x\to 0} f(x) = \lim_{n\to\infty} f(1/4n)$
$= \lim_{n\to\infty} \left((\sin e^n)\, e^{-n^2} + \frac{1}{1+1/n^2}\right) = 0 + 1 = 1$

Q.12 Answer:

12

Solution:

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Given $f''(x)$ is continuous at $x = 0 = \lim_{x\to 0} f''(x) = f''(0) = 4$

Now, $\lim_{x\to 0} \frac{2f(x)-3f(2x)+f(4x)}{x^2}$ $\left[\frac{0}{0}\right.$ form

$= \lim_{x\to 0} \frac{2f'(x)-6f'(2x)+4f'(4x)}{2x}$ $\left[\frac{0}{0}\right.$ form $\left.\right]$

$= \lim_{x\to 0} \frac{2f''(x)-12f''(x)+16f''(4x)}{2}$

[Using L' Hospital Rule successively] $= \frac{2f''(0)-12f''(0)+16f''(0)}{2} = 12$

Q.13 Answer:

1

Solution:

Let $x = a \in Q; f(a) = a$

$f(a^+) = 1 - a$ or a ; $f(a^-) = 1 - a$ or a

continuous at where $1 - a = a \Rightarrow a = 1/2$

$\Rightarrow$ continuous at one point.

Q.14 Answer:

3 only

Solution:

For $x \geq 0 \lim_{x\to 1} f(x) = \lim_{x\to 1} 2 + x = 2 + 1 = 3$

For $x < 0$

$\lim_{x\to 1} f(x) = \lim_{x\to 1} 2 - x = 2 - 1 = 1$

So, $\lim_{x\to 1} f(x)$ does not exist. At $x = 0$ RHL: $\lim_{h\to 0^+} f(0 + h) = \lim_{h\to 0} 2 + h = 2$

LHL : $\lim_{h\to 0^-} f(0 - h) = \lim_{h\to 0} 2 - h = 2$

$f(0) = 2 + 0 = 2.$

So, RHL $=$ LHL $= f(0)$

$\Rightarrow f(x)$ is continuous at $x = 0$ Differentiability at $x = 0$

$$\text{LHD} : \lim_{h\to 0^-} \frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0^-} \frac{2+h-2}{-h}$$

$$= \frac{-h}{h} = -1$$

$$\text{RHD} : \lim_{h\to 0^+} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0^+} \frac{2+h-2}{h} = 1$$

Since LHD $\neq$ RHD So, $f(x)$ is not differentiable at $x = 0.$

Q.15 Answer:

1

Solution:

$\because fx$ is continuous at $x = 3$

$\therefore \lim_{x\to 3} fx = f3$

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$23^2 + 3 = a3^2 + b3 + 1$

$9a + 3b = 20 \quad \ldots 1$

Now, $f'x = \begin{matrix} 4x & ; & x > 3 \\ 2ax + b & ; & x < 3 \end{matrix}$

$\because fx$ **is also differentiable at** $x = 3$

$\therefore 43 = 2a3 + b$

$6a + b = 12 \quad \ldots 2$

From 1 & 2, we get,

$9a = 16 \Rightarrow a = \frac{16}{9}, \quad b = \frac{4}{3}$

$\therefore \frac{a}{b^2} = \frac{16/9}{4/3^2} = 1$

Q.16 Answer:

7.29

Solution:

At $x = 0$**,**

$\text{LHL} = \lim_{h \to 0^+} f0 - h = \lim_{h \to 0^+} \frac{e^{1 - \frac{1}{h}} - a}{e^{-\frac{1}{h}} + 1} = \frac{0 - a}{0 + 1} = -a$

$\text{RHL} = \lim_{h \to 0^+} f0 + h = \lim_{h \to 0^+} \frac{e^{1 + \frac{1}{h}} - a}{e^{\frac{1}{h}} + 1} = \lim_{h \to 0} \frac{e - e^{-\frac{1}{h}} \times a}{1 + e^{-\frac{1}{h}}} = e$

and $f0 = b$

$\because \quad fx$ **is continuous at** $x = 0$

$\therefore \quad -a = e = b \Rightarrow a = -e$

$a^2 = e^2$

Q.17 Answer:

0.5

Solution:

For the function fx **to be continuous at** $x = 2\pi$

$\lim_{x \to 2\pi} fx = f2\pi$

Now, $\lim_{x \to 2\pi} \frac{1 - \cos x}{2\pi - x^2} \frac{\sin^2 x}{\log 1 + 2\pi - x^2} = \lambda$

Putting $x = 2\pi + t$**, we get,**

$\lim_{t \to 0} \frac{1 - \cos t}{t^2} \cdot \frac{\sin^2 t}{\log 1 + t^2} = \lambda$

$\lim_{t \to 0} \frac{1}{2} \cdot \frac{\sin^2 t}{t^2} \cdot \frac{t^2}{\log 1 + t^2} = \lambda$

$\frac{1}{2} \times 1 \times 1 = \lambda$

$\Rightarrow \lambda = \frac{1}{2}$

Q.18 Answer:

differentiable $\forall x \in R - -1, 1$

Solution:

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$$fx = \begin{cases} 1 - x & : \quad x \leq -1 \\ 2 & : \quad -1 < x \leq 1 \\ 1 + x & : \quad x > 1 \end{cases}$$

Continuity at $x = -1$

$f\text{-}1 = 1 - -1 = 2$

$f\text{-}1^{-} = 1 - -1 = 2$

$f\text{-}1^{+} = 2$

$\because f\text{-}1 = f\text{-}1^{-} = f\text{-}1^{+}$

$\therefore$ **continuous at** $x = -1$

$f1 = 2, \quad f1^{-} = 2$

$f1^{+} = 1 + 1 = 2$

$\because f1^{-} = f1 = f1^{+}$

$\therefore$ **continuous at** $x = 1$

For differentiability,

$$f'x = \begin{cases} -1 & x < -1 \\ 0 & -1 < x < 1 \\ 1 & x > 1 \end{cases}$$

at $x = -1$,

$f'\text{-}1^{-} = -1, \quad f\text{-}1^{+} = 0$

$\because f'-1^{-} \neq f'-1^{+} \Rightarrow$ **non-differentiable**

at $x = 1$,

$f'1^{-} = 0, \quad f'1^{+} = 1$

$\because f'1^{-} \neq f'1^{+} \Rightarrow$ **non-differentiable**

Q.19 Answer:

6

Solution:

$f(x) = \left[x^3 - 3\right]$

$(1.26)^3 = 2 - 3 = -1; (1.44)^3 = 3 - 3 = 0$

Similarly, we can check for other points where $f(\mathrm{x})$ **changes values to** $1, 2, 3, 4$. $\therefore$ **Total number of points of discontinuity are ' 6 '**

Q.20 Answer:

1

Solution:

$f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1}$

$= \frac{e^{2x}-1-2x}{x(e^{2x}-1)} = \frac{\left(1+2x+\frac{(2x)^2}{2!}+\ldots\right)-1-2x}{x\left(\left(1+2x+\frac{(2x)^2}{2!}+\ldots\right)-1\right)} = \frac{(2x)^2\left(\frac{1}{2!}+\frac{2x}{3!}+\ldots\right)}{x\cdot(2x)\left(1+\frac{2x}{2!}+\ldots\right)}$

$\therefore \lim_{x\to 0} f(x) = \frac{2\cdot(1/2)}{1} = 1$

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Hence, to make $f(x)$ continuous at $x = 0$,
$f(0)$ must be 1.

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